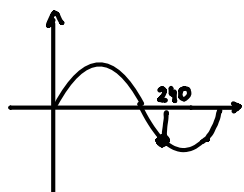


Zadanie 2

Oblicz $\sin 240^\circ \cdot \cos 675^\circ \cdot \frac{\operatorname{tg} \frac{7\pi}{6}}{\operatorname{tg}(-\frac{5\pi}{3})}$.

① Obliczamy $\sin 240^\circ$.



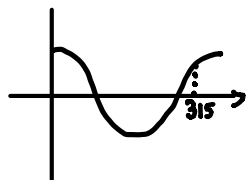
$$\sin 240^\circ = -\frac{\sqrt{3}}{2}$$

$$\sin 240^\circ = \sin(180^\circ + 60^\circ) = -\sin 60^\circ = -\frac{\sqrt{3}}{2}.$$

② Obliczamy $\cos 675^\circ$.

Odejmujemy jeden pełny okres.

$$\cos 675^\circ = \cos(675^\circ - 360^\circ) = \cos 315^\circ$$



$$\cos 315^\circ = \frac{\sqrt{2}}{2}$$

$$\cos 315^\circ = \cos(360^\circ - 45^\circ) = \cos 45^\circ = \frac{\sqrt{2}}{2}.$$

③ Obliczamy $\operatorname{tg} \frac{7\pi}{6}$.

$\frac{7\pi}{6}$ możemy zapisać jako 210° . Wiemy też, że $\operatorname{tg} \alpha = \frac{\sin \alpha}{\cos \alpha}$.

$$\operatorname{tg} 210^\circ = \frac{\sin 210^\circ}{\cos 210^\circ} = \frac{\sin(180^\circ + 30^\circ)}{\cos(180^\circ + 30^\circ)} = \frac{-\sin 30^\circ}{-\cos 30^\circ} = \frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

④ Obliczamy $\operatorname{tg}(-\frac{5\pi}{3})$.

Dodajemy dwa pełne okresy.

$$\operatorname{tg}(-\frac{5\pi}{3} + 2\pi) = \operatorname{tg} \frac{\pi}{3} = \operatorname{tg} 60^\circ = \sqrt{3}$$

⑤ Obliczamy wartość wyrażenia.

$$\sin 240^\circ \cdot \cos 675^\circ \cdot \frac{\operatorname{tg} \frac{7\pi}{6}}{\operatorname{tg}(-\frac{5\pi}{3})} = -\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} \cdot \frac{\frac{\sqrt{3}}{3}}{\sqrt{3}} = \frac{-\sqrt{6}}{4} \cdot \frac{\sqrt{3}}{3} \cdot \frac{1}{\sqrt{3}} = \frac{-\sqrt{6}}{12}$$

Odp.: Wartość wyrażenia wynosi $\frac{-\sqrt{6}}{12}$.

II sposób

Będziemy korzystać ze wzorów redukcyjnych.

① Obliczamy $\sin 240^\circ$.

$$\sin 240^\circ = \sin (180^\circ + 60^\circ) = -\sin 60^\circ = -\frac{\sqrt{3}}{2}$$

② Obliczamy $\cos 675^\circ$.

$$\begin{aligned}\cos 675^\circ &= \cos (360^\circ + 315^\circ) = \cos 315^\circ = \cos (360^\circ - 45^\circ) = \\ &= \cos 45^\circ = \frac{\sqrt{2}}{2}\end{aligned}$$

③ Obliczamy $\operatorname{tg} \frac{7\pi}{6}$.

$$\operatorname{tg} \frac{7\pi}{6} = \operatorname{tg} \left(\pi + \frac{\pi}{6} \right) = \operatorname{tg} \frac{\pi}{6} = \frac{\sqrt{3}}{3}$$

④ Obliczamy $\operatorname{tg} \left(-\frac{5\pi}{3} \right)$.

$$\operatorname{tg} \left(-\frac{5\pi}{3} \right) = \operatorname{tg} \left(-\frac{5\pi}{3} + 2\pi \right) = \operatorname{tg} \frac{2\pi}{3} = \operatorname{tg} \left(\pi - \frac{\pi}{3} \right) = -\operatorname{tg} \frac{\pi}{3} = -\sqrt{3}$$

⑤ Obliczamy wartość wyrażenia.

$$\begin{aligned}&\sin 240^\circ \cdot \cos 675^\circ \cdot \frac{\operatorname{tg} \frac{7\pi}{6}}{\operatorname{tg} \left(-\frac{5\pi}{3} \right)} = \\ &= \frac{-\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} \cdot \frac{\frac{\sqrt{3}}{3}}{-\sqrt{3}} = -\frac{\sqrt{6}}{12}\end{aligned}$$

Odp.: Wartość wyrażenia wynosi $-\frac{\sqrt{6}}{12}$.